THE PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR PRE -MOCK EXAMINATION

CHEMISTRY

Time: 3 hours **March, 2025**

Instructions

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1. This paper consists of section A, B and C with a total of eleven (11) questions

2. Answer all questions in Section A and B. And two questions in section C.

3. Section A carry 16 marks section B carry 54 marks and section C carry 30 marks.

4. Cellular phones and any unauthorized materials are not allowed in examination room.

5. Write your name on every page of your answer sheet.

6. The following constant may be used.

Relative Atomic masses

H=1, C=12, O=16, N=14, Cl=35.5, Cu=64, Pb= 207, Na=23, Ca=40, Mg=24

Avogadro's number 6.02×10^{23}

GMV at S.T. P =$22.4dm3/mol$

1Faraday =96500 coulombs

1 litre =$1dm^3$=$1000cm^3$

SECTION A (16 marks)

Answer all questions in this section

1. For each of the items (i)- (x), choose the correct answer from the given alternatives and write its letter beside the corresponding item number in the booklet provided

(i) What is the IUPAC name for H_2SO_4?

A. Sulphuric (VI) acid
B. Sulphuric (IV) acid
C. Hydrogen sulphate
D. Sulphuri (VII) acid
E. Hydrogen tetra sulphate

(ii) Consider the following fuels which are used for different purposes;

1. Coal 2. Firewood 3. Petrol 4. Charcoal.

Which fuels originate from fossils?

A. 1 and 3 B. 1 and 4 C. 2 and 4 D. 2 and 3 E. 1 and 2

(iii) Glucose dissolves in water to form a uniform mixture

A. Water is solution glucose is solvent and product is solute
B. Water is solute, glucose is solvent and product is solute
C. Water is solvent, glucose is solute and product is solution
D. Water and glucose form immiscible mixture
E. Water and glucose form emulsion

(iv) The reason why white anhydrous copper (II) sulphate turns blue exposed in atmosphere is that it

A. Absorbs water vapour
B. Reacts with oxygen
C. Reacts with carbon dioxide
D. Become dry
E. Release water to the atmosphere

(v) Form Three students in a certain school were doing a scientific procedure about malaria disease in the Ruvuma region. Which of the following scientific procedure was used to accept or reject the hypotheses

A. Conclusion
B. Data interpretation
C. Experimentation
D. Hypothesis formulation
E. Problem identification

(vi) Why sodium hydroxide pellet should be stored in a closed container?

A. Sodium hydroxide is efflorescence.
B. Sodium hydroxide is hygroscopic.

C. Sodium hydroxide is volatile.
D. Sodium hydroxide is deliquescent.
E. Sodium hydroxide is flammable.

(vii) The oxidation number of phosphorus in the radical PO_4^{3-} is;
A. +3 B. +5 C. -3 D. -5 E. 4

(viii) The magnesium salt responsible for permanent hardness of water is;
A. Hydrogen carbonates B. Sulphate C. Nitrate D. Carbonate
E. Chloride

(ix) Organic compounds with general formula RCOOH is known as:
A. Aldehyde
B. Carboxylic acid
C. Alcohol
D. Esters
E. Hydrocarbons

(x) Which of the following are the components needed to start fire
A. Matchbox, firewood and oxygen
B. Match box , firewood and kerosene
C. Oxygen, fuel and firewood
D. Oxygen heat and firewood
E. Hydrogen, chlorine and water

2. Match the materials in **LIST A** with the correct method of preventing it from rusting in **LIST B** by writing the letter of the correct answer in answer sheet provided

LIST A	LIST B
i. Alcohol and water.	A. Layer separation
ii. Cooking oil and water.	B. Solvent extraction
iii. Oil from seeds.	C. Filtration
iv. Sulphur and ion feelings.	D. Magnetization
v. Table salt crystals and ammonium chloride crystals.	E. Chromatography
	F. Sublimation
	G. Fractional distillation
vi. Beans and water.	

SECTION B (54 marks)

Answer all questions in this section

3. (a)A student aimed to prepare gas A by using moderate reactive metal with a dilute acid. By using the information given above, answer the following questions; -

i. What is the name of gas A
ii. What is the chemical test that distinguishes this gas from other gases?
iii. With two reasons, state the correct means of collecting the gas?
iv. Write the balanced chemical equation for the reaction

(b) Mwakipete wanted the above reaction (6) (a) to go fast because wants to utilize in various industrial uses. You as a chemist provide him four means that can be used to make the reaction faster.

4. (a) a form one student has a ballon filled with a certain gas. If the student puts the ballon in the freezer overnight, what do you think will happen to the ballon? Explain

b) How does temperature affect the state of matter.

5. If 20 g of copper(II) oxide reacted with ammonia gas at STP

$$3CuO(s) + 2\,NH_{3(g)} -> 3Cu(s) + N_{2(g)} + H_2O(g)$$

,calculate:

(a) the number of moles of copper(II) oxide used.

(b) the number of moles of ammonia gas used.

(c) the volume of ammonia gas used

6. (a) (i) List physical properties of acid and a base.
(ii)Differentiate between the term's strength of an acid and concentration of an acid.

(b) One day Mr. Mlonga was suffering from heartburn he decided to take in some wood ashes solution. After few hours he got relief

i. Name the process that took place in his stomach.
ii. Explain six applications of the process named above in everyday life.

7 . (a)Give difference between the following pairs of terms as applied in electrolysis:

i. Electrolytes and non-electrolytes
ii. Cathode and anode
iii. Strong electrolytes and weak electrolytes
iv. Electrolytic cell and a battery

(b)A dilute electrolyte is not necessarily weak and a concentrated electrolyte is not

necessarily strong. Explain.

8. (a)Explain the following concepts:
   i. Collision theory
   ii. Effective collision
   iii. Activation energy .

   (b)The manufacture of ammonia by Haber process is an example of a chemical equilibrium of industrial importance.

   $N_2 + 3H_2 \rightleftharpoons 2\ NH_3$

   Describe the effects of temperature and pressure on this reaction

SECTION C 30 Marks.

Answer only 2 questions from this section.

9. Describe the extraction of iron from its ore and write all the chemical equations for the reactions involved in each stage of extraction
10. From the structural formulae of hydrocarbons HCCH2CH3,CH3(CH2)2CH3 and H2CCHCH2CH3.
    a) Which family does each of the compounds belong?
    b) Identify the compound that will decolorize bromine water in the
    i. Dark only
    ii. Presence of UV light only
11. Global warming is one of the major problems in the world. Explain any five effects of global warming